

# NINGHAN ZHONG

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## RESEARCH INTERESTS

My research interests span robot learning and planning: latent world models planning from vision and touch, vision-language-action (VLA) models, and learning and motion planning for contact-rich tasks.

## EDUCATION

### Georgia Institute of Technology

Ph.D. in Robotics

Advisor: Prof. Sriram Vishwanath

Atlanta, GA, USA

Aug 2025 – Present

### University of Waterloo

M.A.Sc. in Electrical and Computer Engineering

Advisor: Prof. Stephen L. Smith

Waterloo, ON, Canada

Aug 2023 – Aug 2025

### University of Illinois Urbana-Champaign

B.S. in Computer Science, *Highest Honors*

Urbana, IL, USA

Aug 2018 – May 2022

## PUBLICATIONS

N. Zhong\*, J.-C. Peng\*, and S. Vishwanath. **GT-VLA: Target-Conditioned Trace Guidance for Generalizable Robotic Manipulation**. *arXiv:2609.31904*, 2026. [Project] [Paper] [arXiv]

N. Zhong\*, S. Caro\*, M. Ramesh, R. Bhatnagar, A. Iskandar, and S. L. Smith. **Bench-Push: Benchmarking Pushing-based Navigation and Manipulation Tasks for Mobile Robots**. *Conference on Robots and Vision (CRV)*, 2026. [Paper] [arXiv] [Code]

R. de Schaetzen, A. Botros, N. Zhong, K. Murrant, R. Gash, and S. L. Smith. **AUTO-IceNav: A Local Navigation Strategy for Autonomous Surface Ships in Broken Ice Fields**. *IEEE Transactions on Robotics (T-RO)*, 2025. [Paper] [arXiv] [Code]

N. Zhong, A. Potenza, and S. L. Smith. **Autonomous Navigation in Ice-Covered Waters with Learned Predictions on Ship-Ice Interactions**. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. [Paper] [arXiv] [Project] [Code]

N. Zhong and K. Hauser. **Attentiveness Map Estimation for Haptic Teleoperation of Mobile Robot Obstacle Avoidance and Approach**. *IEEE Robotics and Automation Letters (RA-L)*, 2024. [Paper] [arXiv]

Z. Huang\*, Y.-J. Mun\*, X. Li<sup>†</sup>, Y. Xie<sup>†</sup>, N. Zhong<sup>†</sup>, W. Liang, J. Geng, T. Chen, and K. Driggs-Campbell. **Hierarchical Intention Tracking for Robust Human-Robot Collaboration in Industrial Assembly Tasks**. *IEEE International Conference on Robotics and Automation (ICRA)*, 2023. [Paper] [arXiv] [Project]

## RESEARCH EXPERIENCE

### Georgia Institute of Technology

*Advised by Prof. Sriram Vishwanath*

Atlanta, GA, USA

Aug 2025 – Present

#### Visuo-Tactile World Models for Manipulation · collaboration with Prof. Glen Chou

world models | visuo-tactile sensing | contact-rich manipulation

ONGOING

- Developing latent world models that fuse vision and tactile to plan contact-rich manipulation.
- Designing latent world model planning framework under tactile safety constraints with uncertainty estimate.

#### GT-VLA: Steering Vision-Language-Action Models with Visual Traces

VLAs | steerability | generalizable manipulation

ARXIV 2026

- Proposed a steerable VLA framework that turns semantic targets from an off-the-shelf generalist VLM into 2D end-effector traces that condition action generation, with no VLM fine-tuning.
- Designed skill-routed Mixture-of-Experts trace and action experts, plus scene-drop, trace-drop, and low-frequency trace-perturbation augmentations that balance trace adherence with geometric flexibility.
- Improved average success on held-out LIBERO suites by +28.0 points over  $\pi_{0.5}$  and +9.9 over AtomicVLA; outperformed baselines on a physical arm in long-horizon pick-and-place, with gains widening with horizon.

## University of Waterloo

*Advised by Prof. Stephen L. Smith*

Waterloo, ON, Canada

Aug 2023 – Aug 2025

### Bench-Push: Benchmarking Pushing-based Navigation and Manipulation

reinforcement learning | motion planning | benchmarking

CRV 2026

- Built an open-source Gymnasium benchmark of four pushing-based tasks for mobile robots (maze, ship-ice navigation, box delivery, area clearing), with MuJoCo environments and real-robot demonstrations.
- Trained and evaluated reinforcement learning (SAC, PPO) and planning baselines (RRT, Spatial Action Maps, GTSP) across all tasks.

### Predictive Planning for Autonomous Ship Navigation in Ice

learned dynamics | path planning | optimization | autonomous navigation

ICRA 2025 | T-RO 2025

- Proposed a deep learning model that predicts the coarse motion of ice floes caused by ship actions through occupancy estimation, and integrated it into a graph-search planner.
- Evaluated in simulation and on a physical dry-land testbed, where the planner reduced collisions compared with existing methods.
- Contributed to AUTO-IceNav, a local navigation strategy for autonomous surface ships in broken ice fields (IEEE T-RO).

## University of Illinois Urbana-Champaign

Urbana, IL, USA

### Attentiveness-Aware Haptic Teleoperation · *advised by Prof. Kris Hauser*

human modeling | obstacle avoidance | haptics

RA-L 2024

- Proposed a real-time model of the operator's spatial attentiveness, combining visual saliency with computational working-memory theory, to adapt haptic feedback for mobile-robot teleoperation.
- Built a perception pipeline that runs environment mapping and attentiveness estimation together; experiments showed shorter task times, less operator control effort, and fewer collisions.

### Intention-Aware Human-Robot Collaboration · *advised by Prof. Katherine Driggs-Campbell*

human-intention prediction | robot perception

ICRA 2023

- Contributed to a robot control framework with hierarchical human-intention tracking for industrial assembly; implemented localization of target assembly parts from the robot's wrist camera.

## INDUSTRY EXPERIENCE

### Golden Ridge Robotics

*Computer Vision and Algorithms Intern*

Chengdu, China

May 2021 – Jul 2021

- Developed an obstacle detection and localization module for low-speed autonomous sightseeing vehicles (C++, ROS, OpenCV, PCL), with under 2% relative error in distance estimation.
- Built a 3D map-annotation tool for marking static obstacles and drivable areas of arbitrary shape in point clouds covering up to 25 km<sup>2</sup>, at decimeter-level accuracy.

## HONORS AND AWARDS

**International Master's Award of Excellence** (\$12,500 CAD)

University of Waterloo, Nov 2023

**Highest Honors** at graduation

UIUC, May 2022

## ACADEMIC SERVICE

**Journal Reviewer:** IEEE Transactions on Robotics (T-RO); Reliability Engineering & System Safety (RESS)

**Conference Reviewer:** IEEE/RSJ IROS 2025; IEEE MRS 2025

## TECHNICAL SKILLS

**Methods:** vision-language-action models, world models, imitation and reinforcement learning, motion planning, perception, human-robot interaction

**Software:** Python, C++, PyTorch, ROS, MuJoCo, Gymnasium, Stable-Baselines3, LIBERO, OpenCV

**Hardware:** Physical robots and sensors such as UR, Franka, YAM, Intel RealSense, GelSight Mini.